



VARDHAMAN COLLEGE OF ENGINEERING (AUTONOMOUS)

Affiliated to JNTUH, Approved by AICTE, Accredited by NAAC with A++ Grade, ISO 9001:2015 Certified
Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

Department of Computer Science and Engineering
B.Tech-CSE-II-I-A (202425)

Module1: Notes

Linear Lists: Applications of Linked Lists, Polynomial Representation, Polynomial addition, Polynomial multiplication, circular Linked Lists, circular queue using linked list, deque using linked list, Priority Queue and its applications.

Facilitator: Dr. Shanthi Makka, Professor, CSE

Applications of Linked Lists

- Polynomial manipulation
- Dynamic memory allocation
- Implementation of stacks and queues
- Graph adjacency list representation
- Hash tables (chaining for collision handling)
- Navigation systems (undo/redo functionality)
- Memory management (garbage collection)
- Sparse matrix representation
- Music and video playlist management
- Undo/redo operations in software
- File system directories
- Operating system task scheduling

Polynomial Representation using Linked List

Structure of Node

Coefficient	Degree	Address of next node
-------------	--------	----------------------

- Node contains three values:
- **Coefficient:** The numeric value.
- **Exponent or degree:** The power of the variable x.
- **Address:** The address of the next node of the linked list.

Example1:



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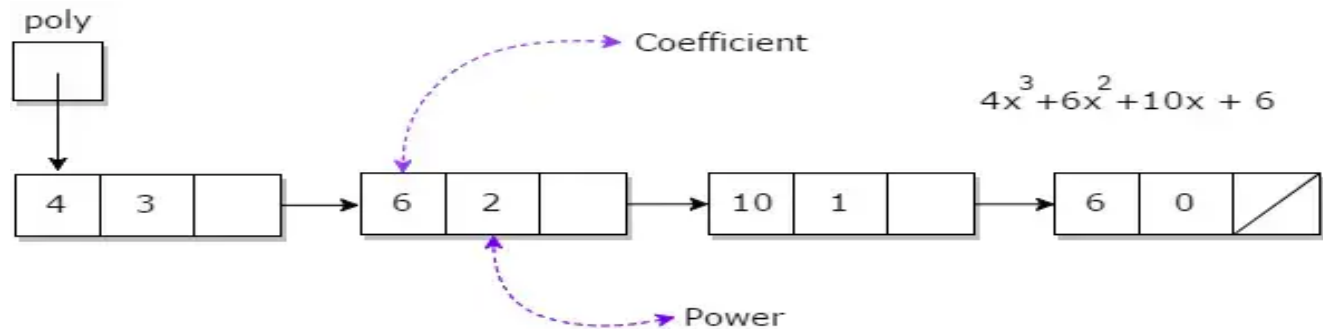
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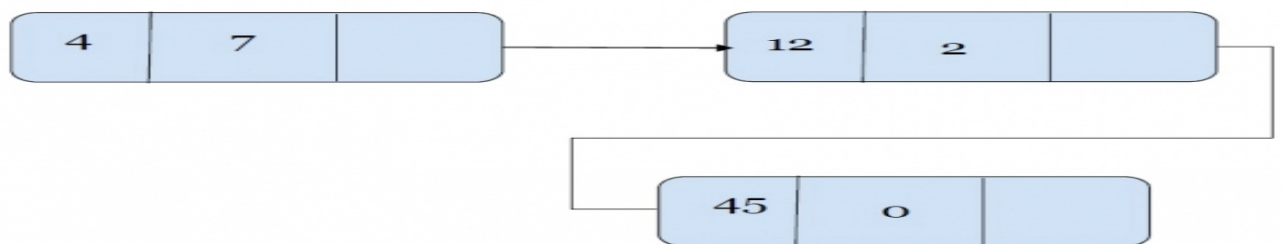
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Example2: Polynomial : $4x^7 + 12x^2 + 45$



Structure Definition:

```
struct Node {  
    int coefficient;  
    int exponent;  
    struct Node *next;  
}
```

Polynomial addition

- To add two polynomials, we traverse the list P and Q. We take corresponding terms of the list P and Q and compare their exponents.
- If the two exponents are equal, the coefficients are added to create a new coefficient. If the new coefficient is equal to 0, then the term is dropped, and if it is not zero, it is inserted at the end of the new linked list containing the resulting polynomial.



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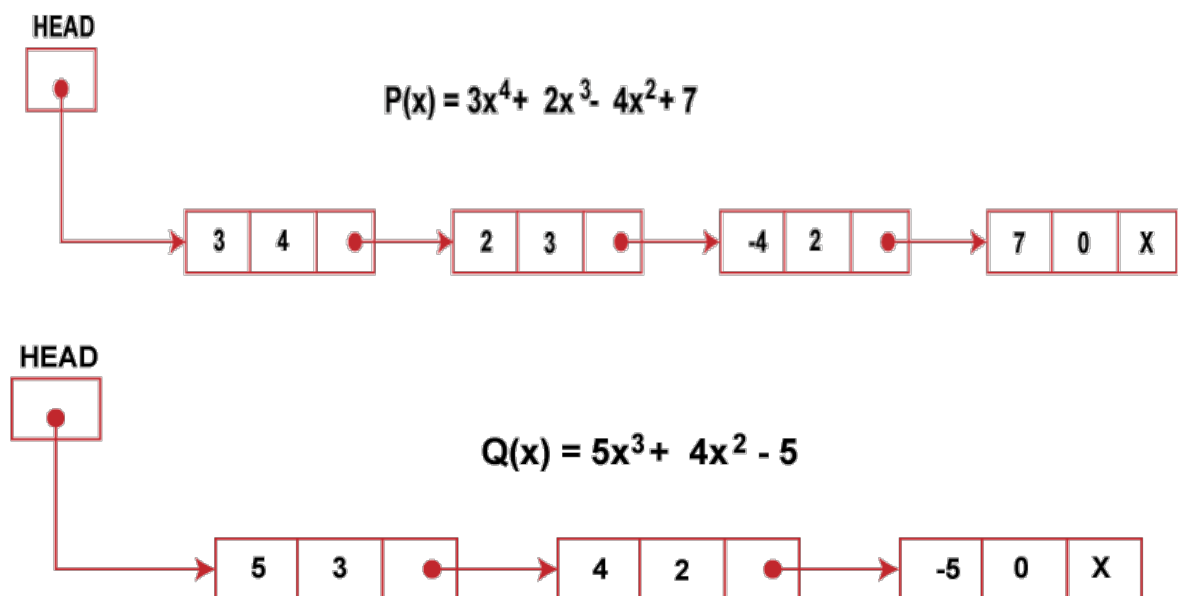
- If one of the exponents is larger than the other, the corresponding term is immediately placed into the new linked list, and the term with the smaller exponent is held to be compared with the next term from the other list.
- If one list ends before the other, the rest of the terms of the longer list is inserted at the end of the new linked list containing the resulting polynomial.

Let us consider an example an example to show how the addition of two polynomials is performed,

$$P(x) = 3x^4 + 2x^3 - 4x^2 + 7$$

$$Q(x) = 5x^3 + 4x^2 - 5$$

These polynomials are represented using a linked list in order of decreasing exponents as follows:



To generate a new linked list for the resulting polynomials that is formed on the addition of given polynomials P(x) and Q(x), we perform the following steps,

Traverse the two lists P and Q and examine all the nodes.



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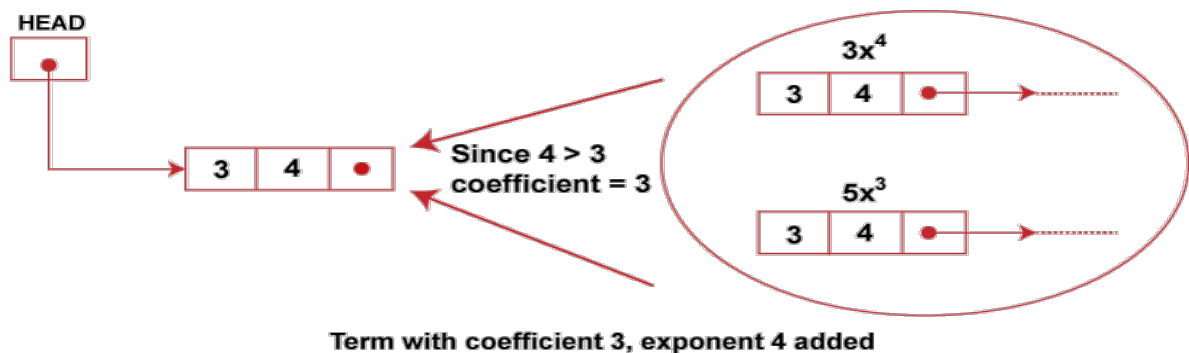
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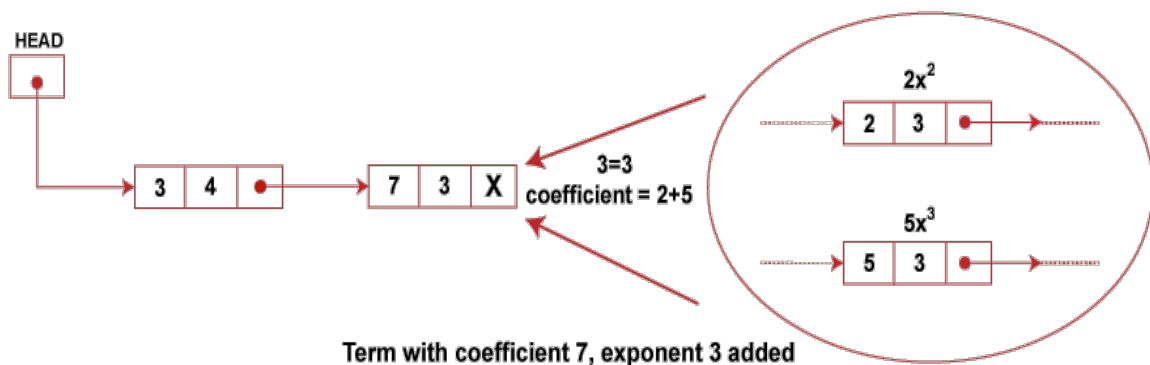
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We compare the exponents of the corresponding terms of two polynomials. The first term of polynomials P and Q contain exponents 4 and 3, respectively. Since the exponent of the first term of the polynomial P is greater than the other polynomial Q, the term having a larger exponent is inserted into the new list. The new list initially looks as shown below:



We then compare the exponent of the next term of the list P with the exponents of the present term of list Q. Since the two exponents are equal, so their coefficients are added and appended to the new list as follows:



Then we move to the next term of P and Q lists and compare their exponents. Since exponents of both these terms are equal and after addition of their coefficients, we get 0, so the term is



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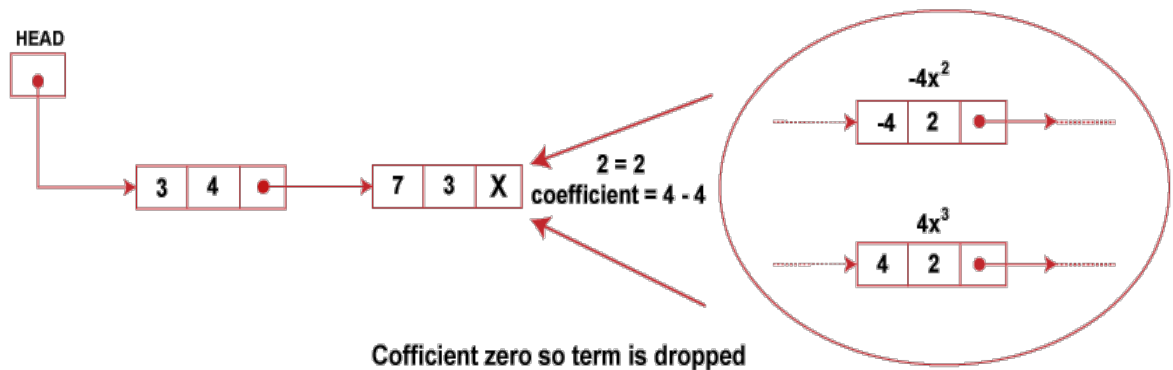
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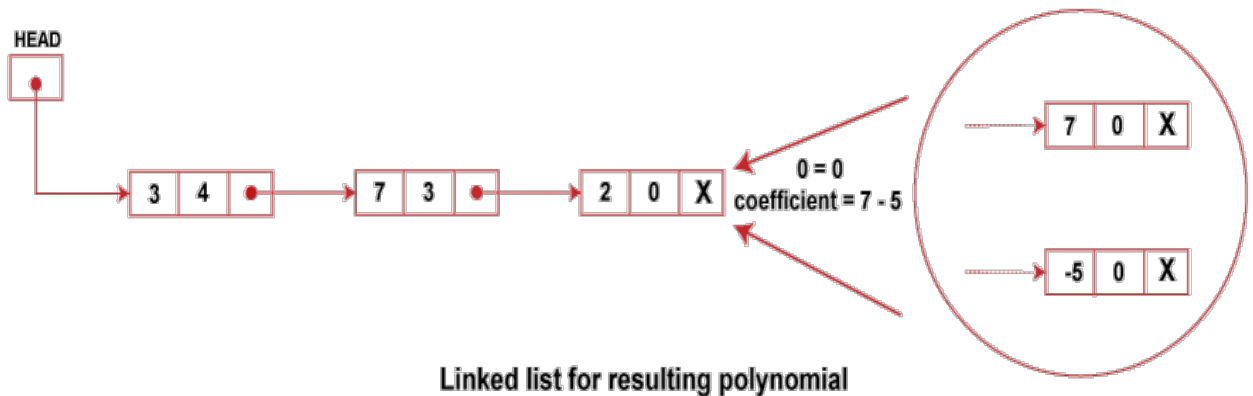
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dropped, and no node is appended to the new list after this,



Moving to the next term of the two lists, P and Q, we find that the corresponding terms have the same exponents equal to 0. We add their coefficients and append them to the new list for the resulting polynomial as shown below:



C Program

```
#include<stdio.h>
#include<stdlib.h>
```

```
struct Node{
    int coeff;
    int pow;
    struct Node *next;
};
```



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```
void create_node(int x, int y, struct Node temp){
    struct Node *r, *z;
    z = *temp;
    if(z == NULL){
        r = (struct Node*)malloc(sizeof(struct Node));
        r->coeff = x;
        r->pow = y;
        *temp = r;
        r->next = (struct Node*)malloc(sizeof(struct Node));
        r = r->next;
        r->next = NULL;
    } else {
        r->coeff = x;
        r->pow = y;
        r->next = (struct Node*)malloc(sizeof(struct Node));
        r = r->next;
        r->next = NULL;
    }
}

void polyadd(struct Node *p1, struct Node *p2, struct Node *result){
    while(p1->next && p2->next){
        if(p1->pow > p2->pow){
            result->pow = p1->pow;
            result->coeff = p1->coeff;
            p1 = p1->next;
        }
        else if(p1->pow < p2->pow){
            result->pow = p2->pow;
            result->coeff = p2->coeff;
            p2 = p2->next;
        }
        else {
            result->pow = p1->pow;
            result->coeff = p1->coeff+p2->coeff;
            p1 = p1->next;
            p2 = p2->next;
        }
        result->next = (struct Node *)malloc(sizeof(struct Node));
        result = result->next;
    }
}
```



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```
result->next = NULL;
}
while(p1->next || p2->next){
    if(p1->next){
        result->pow = p1->pow;
        result->coeff = p1->coeff;
        p1 = p1->next;
    }
    if(p2->next){
        result->pow = p2->pow;
        result->coeff = p2->coeff;
        p2 = p2->next;
    }
    result->next = (struct Node *)malloc(sizeof(struct Node));
    result = result->next;
    result->next = NULL;
}
}
void printpoly(struct Node *node){
    while(node->next != NULL){
        printf("%dx^%d", node->coeff, node->pow);
        node = node->next;
        if(node->next != NULL)
            printf(" + ");
    }
}
int main(){
    struct Node *p1 = NULL, *p2 = NULL, *result = NULL;
    create_node(41,7,&p1);
    create_node(12,5,&p1);
    create_node(65,0,&p1);
    create_node(21,5,&p2);
    create_node(15,2,&p2);
    printf("polynomial 1: ");
    printpoly(p1);
    printf("\npolynomial 2: ");
    printpoly(p2);
    result = (struct Node *)malloc(sizeof(struct Node));
```




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```
polyadd(p1, p2, result);  
printf("\npolynomial after adding p1 and p2 : ");  
printpoly(result);  
return 0;  
}
```

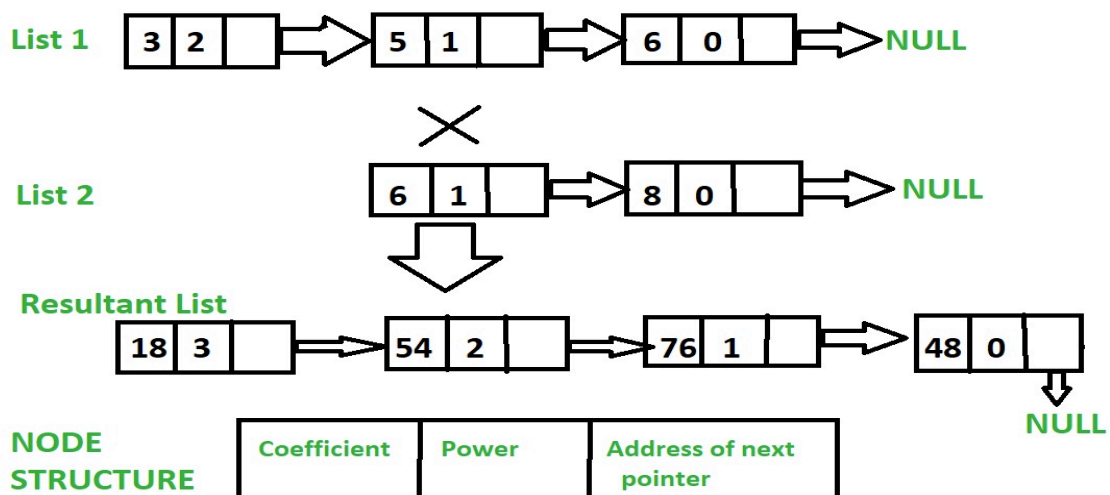
Time Complexity

The time required to add both lists is $O(n + m)$, where n and m represent the length of both lists respectively.

Polynomial Multiplication

Approach:

- In this approach we will multiply the 2nd polynomial with each term of 1st polynomial.
- Store the multiplied value in a new linked list.
- Then we will add the coefficients of elements having the same power in resultant polynomial.



```
#include <stdio.h>  
#include <stdlib.h>  
// Define a structure for a node (term) in the polynomial linked list
```




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```
struct Node {
    int coefficient;
    int exponent;
    struct Node* next;
};

// Function to create a new node
struct Node* createNode(int coeff, int exp) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->coefficient = coeff;
    newNode->exponent = exp;
    newNode->next = NULL;
    return newNode;
}

// Function to insert a node into the polynomial linked list
void insertTerm(struct Node* poly, int coeff, int exp) {
    struct Node* newNode = createNode(coeff, exp);
    if (*poly == NULL) {
        *poly = newNode;
    } else {
        struct Node* temp = *poly;
        while (temp->next != NULL) {
            temp = temp->next;
        }
        temp->next = newNode;
    }
}

// Function to add two polynomials and store the result
struct Node* addPolynomials(struct Node* poly1, struct Node* poly2) {
    struct Node* result = NULL;
    while (poly1 != NULL && poly2 != NULL) {
        if (poly1->exponent > poly2->exponent) {
            insertTerm(&result, poly1->coefficient, poly1->exponent);
            poly1 = poly1->next;
        } else if (poly1->exponent < poly2->exponent) {
            insertTerm(&result, poly2->coefficient, poly2->exponent);
            poly2 = poly2->next;
        } else {
            // Exponents are equal, add coefficients
            int sum = poly1->coefficient + poly2->coefficient;
            if (sum != 0) {
                insertTerm(&result, sum, poly1->exponent);
            }
            poly1 = poly1->next;
            poly2 = poly2->next;
        }
    }
    // Add remaining terms from poly1 or poly2
    while (poly1 != NULL) {
        insertTerm(&result, poly1->coefficient, poly1->exponent);
        poly1 = poly1->next;
    }
    while (poly2 != NULL) {
        insertTerm(&result, poly2->coefficient, poly2->exponent);
        poly2 = poly2->next;
    }
    return result;
}
```



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```
int sumCoeff = poly1>coefficient + poly2>coefficient;
if (sumCoeff != 0) {
    insertTerm(&result, sumCoeff, poly1>exponent);
}
poly1 = poly1>next;
poly2 = poly2>next;
}
}
// Add the remaining terms of poly1 (if any)
while (poly1 != NULL) {
    insertTerm(&result, poly1>coefficient, poly1>exponent);
    poly1 = poly1>next;
}
// Add the remaining terms of poly2 (if any)
while (poly2 != NULL) {
    insertTerm(&result, poly2>coefficient, poly2>exponent);
    poly2 = poly2>next;
}
return result;
}

// Function to multiply two polynomials and store the result
struct Node* multiplyPolynomials(struct Node* poly1, struct Node* poly2) {
    struct Node* result = NULL;

    // Traverse the first polynomial
    while (poly1 != NULL) {
        struct Node* tempPoly2 = poly2;
        struct Node* tempResult = NULL;

        // Traverse the second polynomial
        while (tempPoly2 != NULL) {
            int coeff = poly1>coefficient * tempPoly2>coefficient;
            int exp = poly1>exponent + tempPoly2>exponent;
            insertTerm(&tempResult, coeff, exp);
            tempPoly2 = tempPoly2>next;
        }
        // Add tempResult to the result polynomial
```



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```
    result = addPolynomials(result, tempResult);
    poly1 = poly1->next;
}
return result;
}

// Function to display a polynomial
void displayPolynomial(struct Node* poly) {
    while (poly != NULL) {
        printf("%dx^%d", poly->coefficient, poly->exponent);
        poly = poly->next;
        if (poly != NULL) {
            printf(" + ");
        }
    }
    printf("\n");
}
```

// Main function

```
int main() {
    struct Node* poly1 = NULL;
    struct Node* poly2 = NULL;
    struct Node* result = NULL;

    // First polynomial: 3x^3 + 5x^1 + 6
    insertTerm(&poly1, 3, 3);
    insertTerm(&poly1, 5, 1);
    insertTerm(&poly1, 6, 0);

    // Second polynomial: 6x^2 + 4x^1 + 2
    insertTerm(&poly2, 6, 2);
    insertTerm(&poly2, 4, 1);
    insertTerm(&poly2, 2, 0);

    printf("First Polynomial: ");
    displayPolynomial(poly1);

    printf("Second Polynomial: ");
```



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displayPolynomial(poly2);

```
// Multiply the two polynomials  
result = multiplyPolynomials(poly1, poly2);
```

```
// Display the result  
printf("Resultant Polynomial after multiplication: ");  
displayPolynomial(result);  
return 0;  
}
```

Time Complexity

The time complexity for the abovediscussed approach is $O(n*m)$, where n is the number of nodes in the first polynomial and m is the number of nodes in the second polynomial because we used two nested loops.

Circular Linked List

- In a circular linked list, the last node of the list contains a pointer to the first node of the list.
- We traverse a circular singly linked list until we reach the same node where we started.
- The circular singly linked list has no beginning and no ending.
- There is no null value present in the next part of any of the nodes.

The following image shows a circular singly linked list.



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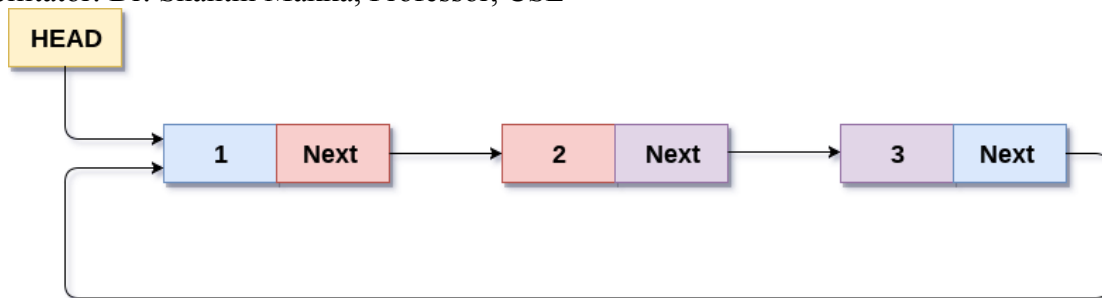
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Circular Singly Linked List

- Circular linked list are mostly used in task maintenance in operating systems.
- There are many examples where circular linked list are being used in computer science including browser surfing where a record of pages visited in the past by the user, is maintained in the form of circular linked lists and can be accessed again on clicking the previous button.

Memory Representation of circular linked list

In the following image, memory representation of a circular linked list containing marks of a student in 4 subjects. However, the image shows a glimpse of how the circular list is being stored in the memory. The start or head of the list is pointing to the element with the index 1 and containing 13 marks in the data part and 4 in the next part. Which means that it is linked with the node that is being stored at 4th index of the list.

However, due to the fact that we are considering circular linked list in the memory therefore the last node of the list contains the address of the first node of the list.



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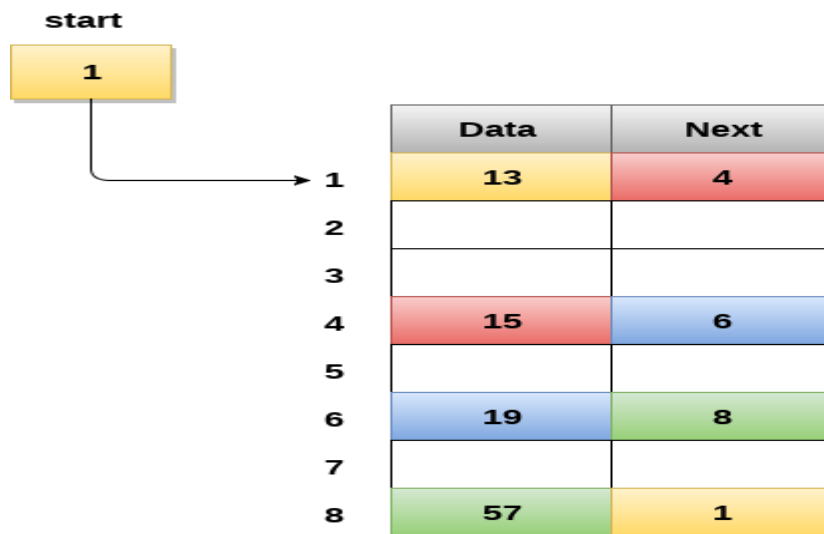
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Memory Representation of a circular linked list

We can also have more than one number of linked lists in the memory with the different start pointers pointing to the different start nodes in the list. The last node is identified by its next part, which contains the address of the start node of the list. We

must be able to identify the last node of any linked list so that we can find out the number of iterations, which need to be performed while traversing the list.

Operations on Circular Singly linked list

Insertion

SN	Operation	Description
1	Insertion at beginning	Adding a node into circular singly linked list at the beginning.



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2	Insertion at the end	Adding a node into circular singly linked list at the end.
3	Insertion at Random Position	Adding a node into circular singly linked list at the specified position.

Deletion & Traversing

SN	Operation	Description
1	Deletion at beginning	Removing the node from circular singly linked list at the beginning.
2	Deletion at the end	Removing the node from circular singly linked list at the end.
3	Deletion from random position	Removing the node from circular singly linked list from any random position given.
3	Searching	Compare each element of the node with the given item and return the location at which the item is present in the list otherwise return null.
4	Traversing	Visiting each element of the list at least once in order to perform some specific operation.

Menu driven program in C implementing all operations on circular singly linked list

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};
struct node *head;
```




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```
void beginsert ();
void lastinsert ();
void randominsert();
void begin_delete();
void last_delete();
void random_delete();
void display();
void search();
void main ()
{
    int choice =0;
    while(choice != 7)
    {
        printf("\n*Main Menu*\n");
        printf("\nChoose one option from the following list ...\n");
        printf("\n===== \n");
        printf("\n1.Insert in begining\n2.Insert at last\n3.Delete from Beginning\n4.Delete from last\n5.
Search for      an element\n6.Show\n7.Exit\n");
        printf("\nEnter your choice?\n");
        scanf("\n%d",&choice);
        switch(choice)
        {
            case 1:
                beginsert();
                break;
            case 2:
                lastinsert();
                break;
            case 3:
                begin_delete();
                break;
            case 4:
                last_delete();
                break;
            case 5:
                search();
                break;
            case 6:
```



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Department of Computer Science and Engineering
B.Tech-CSE-II-I-A (202425)

Module1: Notes

Linear Lists: Applications of Linked Lists, Polynomial Representation, Polynomial addition, Polynomial multiplication, circular Linked Lists, circular queue using linked list, deque using linked list, Priority Queue and its applications.

Facilitator: Dr. Shanthi Makka, Professor, CSE

```
        display();
        break;
    case 7:
        exit(0);
        break;
    default:
        printf("Please enter valid choice..");
    }
}
}
void beginsert()
{
    struct node *ptr,*temp;
    int item;
    ptr = (struct node *)malloc(sizeof(struct node));
    if(ptr == NULL)
    {
        printf("\nOVERFLOW");
    }
    else
    {
        printf("\nEnter the node data?");
        scanf("%d",&item);
        ptr->data = item;
        if(head == NULL)
        {
            head = ptr;
            ptr->next = head;
        }
        else
        {
            temp = head;
            while(temp->next != head)
                temp = temp->next;
            ptr->next = head;
            temp->next = ptr;
            head = ptr;
        }
    }
}
```



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```
printf("\nnode inserted\n");
}
}
void lastinsert()
{
    struct node *ptr,*temp;
    int item;
    ptr = (struct node *)malloc(sizeof(struct node));
    if(ptr == NULL)
    {
        printf("\nOVERFLOW\n");
    }
    else
    {
        printf("\nEnter Data?");
        scanf("%d",&item);
        ptr->data = item;
        if(head == NULL)
        {
            head = ptr;
            ptr->next = head;
        }
        else
        {
            temp = head;
            while(temp->next != head)
            {
                temp = temp->next;
            }
            temp->next = ptr;
            ptr->next = head;
        }
        printf("\nnode inserted\n");
    }
}

void begin_delete()
{
```



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```
struct node *ptr;
if(head == NULL)
{
    printf("\nUNDERFLOW");
}
else if(head->next == head)
{
    head = NULL;
    free(head);
    printf("\nnode deleted\n");
}

else
{
    ptr = head;
    while(ptr->next != head)
        ptr = ptr->next;
    ptr->next = head->next;
    free(head);
    head = ptr->next;
    printf("\nnode deleted\n");
}
}
void last_delete()
{
    struct node *ptr, *preptr;
    if(head==NULL)
    {
        printf("\nUNDERFLOW");
    }
    else if (head->next == head)
    {
        head = NULL;
        free(head);
        printf("\nnode deleted\n");
    }
    else
    {
        ptr = head;
```



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```
while(ptr >next != head)
{
    preptr=ptr;
    ptr = ptr>next;
}
preptr>next = ptr > next;
free(ptr);
printf("\nnode deleted\n");
}
```

```
void search()
{
    struct node *ptr;
    int item,i=0,flag=1;
    ptr = head;
    if(ptr == NULL)
    {
        printf("\nEmpty List\n");
    }
    else
    {
        printf("\nEnter item which you want to search?\n");
        scanf("%d",&item);
        if(head->data == item)
        {
            printf("item found at location %d",i+1);
            flag=0;
        }
        else
        {
            while (ptr->next != head)
            {
                if(ptr->data == item)
                {
                    printf("item found at location %d ",i+1);
                    flag=0;
                    break;
                }
            }
        }
    }
}
```



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```
    }
    else
    {
        flag=1;
    }
    i++;
    ptr = ptr > next;
}
}
if(flag != 0)
{
    printf("Item not found\n");
}
}
```

```
void display()
{
    struct node *ptr;
    ptr=head;
    if(head == NULL)
    {
        printf("\nnothing to print");
    }
    else
    {
        printf("\n printing values ... \n");

        while(ptr > next != head)
        {
            printf("%d\n", ptr > data);
            ptr = ptr > next;
        }
        printf("%d\n", ptr > data);
    }
}
```

Circular Linked List Applications



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- It is used in multiplayer games to give a chance to each player to play the game.
- Multiple running applications can be placed in a circular linked list on an operating system. The OS keeps on iterating over these applications.

Operation	Time Complexity
Search	$O(n)$
Insertion	$O(1)$ to $O(n)$
Deletion	$O(1)$ to $O(n)$
Traversal	$O(n)$

Circular Queue using linked list

Operations on Circular Queue:

1. enqueue(data)
 - Create a struct node type node.
 - Insert the given data in the new node data section and NULL in address section.
 - If Queue is empty then initialize front and rear from new node.
 - Queue is not empty then initialize rear next and rear from new node.
 - New node next initialize from front
2. dequeue()
 - Check if queue is empty or not.
 - If queue is empty then dequeue is not possible.
 - Else Initialize temp from front.
 - If front is equal to the rear then initialize front and rear from null.
 - Print data of temp and free temp memory.
 - If there is more than one node in Queue then make front next to front then initialize rear next from front.
 - Print temp and free temp.
3. print() or traversal()



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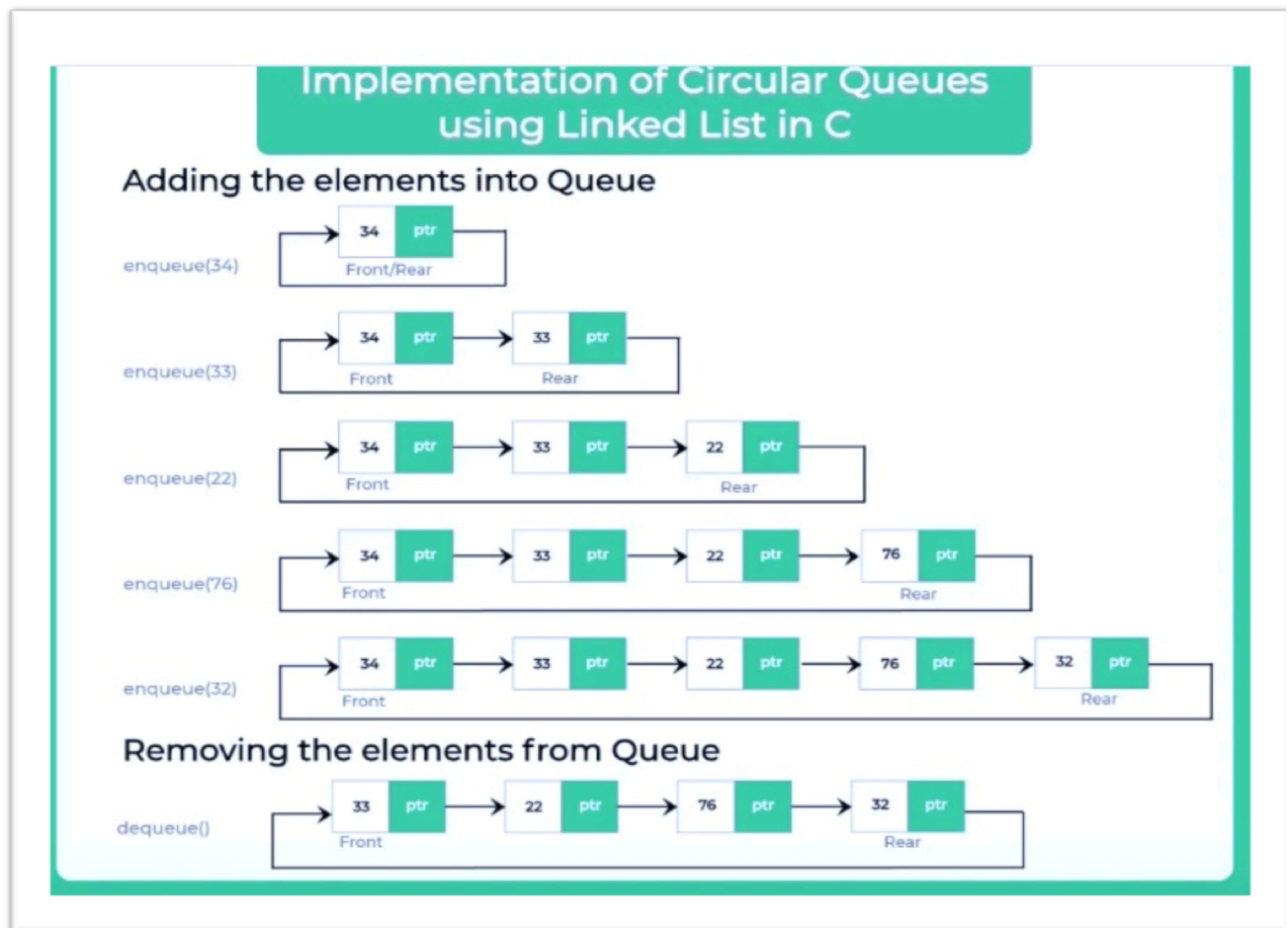
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- Check if there is some data in the queue or not.
- If the queue is empty print “No data in the queue.”
- Else define a node pointer and initialize it with front.
- Print data of node pointer until the next of node pointer becomes NULL.



C Program

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
```



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```
int data;  
struct node *next;  
};
```

```
struct node *f = NULL;  
struct node *r = NULL;
```

```
void enqueue (int d)                //Insert elements in Queue  
{  
    struct node *n;  
    n = (struct node *) malloc (sizeof (struct node));  
    n->data = d;  
    n->next = NULL;  
    if ((r == NULL) && (f == NULL))  
    {  
        f = r = n;  
        r->next = f;  
    }  
    else  
    {  
        r->next = n;  
        r = n;  
        n->next = f;  
    }  
}
```

```
void dequeue ()                    // Delete an element from Queue  
{  
    struct node *t;  
    t = f;  
    if ((f == NULL) && (r == NULL))  
        printf ("\nQueue is Empty");  
    else if (f == r)  
    {  
        f = r = NULL;  
        free (t);  
    }  
}
```



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```
else
{
    f = f->next;
    r->next = f;
    free (t);
}

}

void display ()
{
    // Print the elements of Queue
    struct node *t;
    t = f;
    if ((f == NULL) && (r == NULL))
        printf ("\nQueue is Empty");
    else
    {
        do
        {
            printf (" %d", t->data);
            t = t->next;
        }
        while (t != f);
    }
}

int main ()
{
    enqueue (34);
    enqueue (22);
    enqueue (75);
    enqueue (99);
    enqueue (27);
    printf ("Circular Queue: ");
    display ();
    printf ("\n");
}
```



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```
dequeue ();  
printf ("Circular Queue After dequeue: ");  
display ();  
return 0;  
}
```

Time complexity

- Search: $O(n)$
- Insertion: $O(1)$ to $O(n)$
- Deletion: $O(1)$ to $O(n)$
- Traversal: $O(n)$

DEQUE using Linked List

- The deque stands for Double Ended Queue.
- Deque is a linear data structure where the insertion and deletion operations are performed from both ends.



Representation of deque

Types of deque

There are two types of deque

- Input restricted queue
- Output restricted queue

Input restricted DeQueue

In input restricted queue, insertion operation can be performed at only one end, while deletion can be performed from both ends.



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input restricted double ended queue

Output restricted DeQueue

In outputrestricted queue, deletion operation can be performed at only one end, while insertion can be performed from both ends.



Output restricted double ended queue

Operations on DEQUE

There are the following operations that can be applied on a deque

- Insertion at front
- Insertion at rear
- Deletion at front
- Deletion at rear

Doubly Linked List Representation of Deque :

For implementing deque, we need to keep track of two pointers, front and rear. We enqueue (push) an item at the rear or the front end of deque and dequeue(pop) an item from both rear and front end.

Working :

Declare two pointers front and rear of type Node, where Node represents the structure of a node of a doubly linked list. Initialize both of them with value NULL.



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Insertion at Front end :

1. Allocate space for a newNode of doubly linked list.
2. IF newNode == NULL, then
3. print "Overflow"
4. ELSE
5. IF front == NULL, then
6. rear = front = newNode
7. ELSE
8. newNode>next = front
9. front>prev = newNode
10. front = newNode

Insertion at Rear end :

1. Allocate space for a newNode of doubly linked list.
2. IF newNode == NULL, then
3. print "Overflow"
4. ELSE
5. IF rear == NULL, then
6. front = rear = newNode
7. ELSE
8. newNode>prev = rear
9. rear>next = newNode
10. rear = newNode

Deletion from Front end :

1. IF front == NULL
2. print "Underflow"
3. ELSE
4. Initialize temp = front
5. front = front>next
6. IF front == NULL
7. rear = NULL
8. ELSE
9. front>prev = NULL
10. Deallocate space for temp



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Deletion from Rear end :

1. IF front == NULL
2. print "Underflow"
3. ELSE
4. Initialize temp = rear
5. rear = rear->prev
6. IF rear == NULL
7. front = NULL
8. ELSE
9. rear->next = NULL
10. Deallocate space for temp

C Program to implement DEQUE using Doubly Linked List

```
#include<stdio.h>

#include<malloc.h>

struct node
{
    int data;

    struct node *prev,*next;
}*front=NULL,*rear=NULL,*newnode,*p,*q; int size=0;

void create();
void display();
void InsertFront(int);
void InsertRear(int);
void DeleteFront();
void DeleteRear();
void getFront();
void getRear();
```




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```
void qsize();
```

```
void erase();
```

```
void main()
```

```
{
```

```
    create(); display();
```

```
        InsertFront(23);
```

```
        InsertRear(34);
```

```
    display(); DeleteFront();
```

```
    DeleteRear(); display();
```

```
    getRear(); getFront(); qsize();
```

```
    erase();
```

```
    display();
```

```
}
```

```
    void create()
```

```
    {
```

```
        int n,i;
```

```
        printf("Enter the no of nodes\n");
```

```
        scanf("%d",&n);
```

```
        for(i=1;i<=n;i++)
```

```
        {
```

```
            newnode=(struct node *) malloc(sizeof(struct node));
```

```
            printf("Enter the data\n");
```

```
            scanf("%d",&newnode->data); newnode->next=NULL;
```

```
            newnode->prev=NULL;
```

```
                if(front==NULL)
```

```
                    front=rear=newn
```

```
                    ode; else
```

```
                {
```

```
                    rear->next=newnode;
```

```
                    newnode->prev=rear;
```

```
                    rear=newnode;
```

```
                }
```

```
                size++;
```

```
            }
```

```
    }
```

```
    void display()
```




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```
{
    q=front;p=rear; printf("The elements are ");
    while(q)
    {
        printf("%3d",q->data); q=q->next;
    }
    printf("\n\nThe elements in reverse order are "); while(p)
    {
        printf("%3d",p->data); p=p->prev;
    }
    printf("\n\n");
}

void InsertFront(int x)
{
    newnode=(struct node
    *)malloc(sizeof(struct node));
    newnode->data=x;
    newnode->next=NULL; newnode->prev=NULL;
    if (front == NULL)
        rear = front = newnode; else
    {
        newnode->next = front; front
        >prev = newnode;
        front = newnode;
    }
    size++;
}

void InsertRear(int x)
{
    newnode=(struct node *)malloc(sizeof(struct node));
    newnode->data=x;
    newnode->next=NULL;
    newnode->prev=NULL;
    if (front == NULL)
        rear = front = newnode;
    else
    {
```



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```
newnode->prev = rear; rear
    >next = newnode; rear =
    newnode;
}
size++;
}
void DeleteFront()
{
    if(front==NULL)
        printf("Deque is
        empty\n");
    else
    {
        p=front; front=front->next;
        if(front==NULL) rear=NULL;
        else
            front->prev=NULL;
        free(p); size;
    }
}
void DeleteRear()
{
    if(front==NULL)
        printf("Deque is empty\n");
    else
    {
        p=rear; rear=rear->prev;
        if(rear==NULL) front=NULL; else
            rear->next=NULL;
        free(p); size;
    }
}
void getRear()
{
    if (front==NULL) printf("Queue is empty\n");
    else
        printf("The rear element is %d \n", rear->data);
}
```



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```
}  
void getFront()  
{  
    if (front==NULL) printf("Queue is empty\n");  
    else  
        printf("The front element is %d\n ", front->data);  
}  
void qsize()  
{  
    printf("The no of nodes in queue are %d\n",size);  
}  
void erase()  
{  
    rear = NULL;  
    while (front != NULL)  
    {  
        p = front;  
        front = front->next; free(p);  
    }  
    size = 0;  
}
```

Time Complexity : Time complexity of operations like insertFront(), insertRear(), deleteFront(), deleteRear() is $O(1)$.

Priority Queue and It's Applications

A priority queue is a special type of queue in which each element is associated with a priority value. And, elements are served on the basis of their priority. That is, higher priority elements are served first. However, if elements with the same priority occur, they are served according to their order in the queue.

Characteristics of a Priority queue

A priority queue is an extension of a queue that contains the following characteristics:

- Every element in a priority queue has some priority associated with it.
- An element with the higher priority will be deleted before the deletion of the lesser priority.



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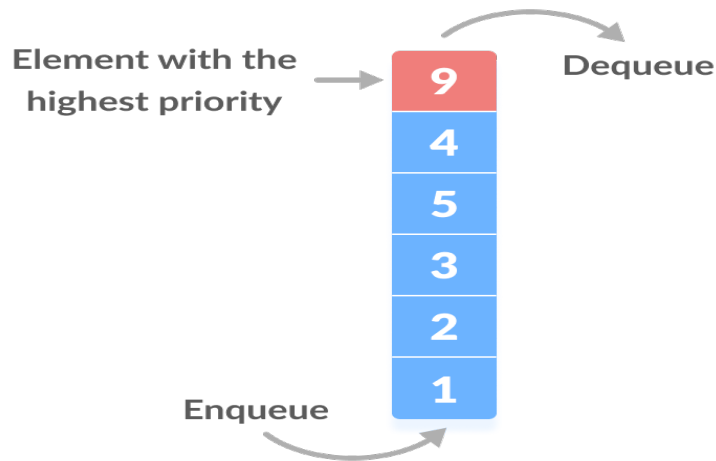
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- If two elements in a priority queue have the same priority, they will be arranged using the FIFO principle.



Operations	peek	insert	delete
Linked List	$O(1)$	$O(n)$	$O(1)$

There are two types of priority queue:

Ascending order priority queue: In ascending order priority queue, a lower priority number is given as a higher priority in a priority. For example, we take the numbers from 1 to 5 arranged in an ascending order like 1,2,3,4,5; therefore, the smallest number, i.e., 1 is given as the highest priority in a priority queue.



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Descending order priority queue: In descending order priority queue, a higher priority number is given as a higher priority in a priority. For example, we take the numbers from 1 to 5 arranged in descending order like 5, 4, 3, 2, 1; therefore, the largest number, i.e., 5 is given as the highest priority in a priority queue.



Representation of priority queue

We will create the priority queue by using the list given below in which **INFO** list contains the data elements, **PRN** list contains the priority numbers of each data element available in the **INFO** list, and **LINK** basically contains the address of the next node.



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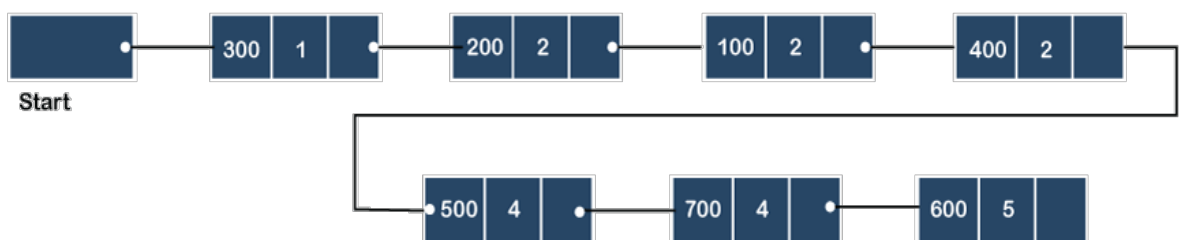
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// C code to implement Priority Queue using Linked List

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// Node
```

```
typedef struct node {
    int data;
```

```
    // Lower values indicate higher priority
    int priority;
```

```
    struct node* next;
```

```
} Node;
```

```
// Function to Create A New Node
```

```
Node* newNode(int d, int p)
```

```
{
```

```
    Node* temp = (Node*)malloc(sizeof(Node));
```

```
    temp->data = d;
```

```
    temp->priority = p;
```

```
    temp->next = NULL;
```

```
    return temp;
```

```
}
```

```
// Return the value at head
```



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```
int peek(Node* head)
```

```
{  
    return head->data;  
}
```

```
// Removes the element with the
```

```
// highest priority from the list
```

```
Node* pop(Node* head)
```

```
{  
    Node* temp = head;  
    head = head->next;  
    free(temp);  
    return head;  
}
```

```
// Function to push according to priority
```

```
Node* push(Node* head, int d, int p)
```

```
{  
    Node* start = head;  
  
    // Create new Node  
    Node* temp = newNode(d, p);  
  
    // Special Case: The head of list has lesser  
    // priority than new node. So insert new  
    // node before head node and change head node.  
    if (head->priority > p) {
```

```
        // Insert New Node before head  
        temp->next = head;  
        head = temp;  
    }  
    else {
```

```
        // Traverse the list and find a  
        // position to insert new node  
        while (start->next != NULL &&  
            start->next->priority < p) {
```




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```
start = start->next;
}

// Either at the ends of the list
// or at required position
temp->next = start->next;
start->next = temp;
}
return head;
}

// Function to check is list is empty
int isEmpty(Node* head)
{
    return (head == NULL);
}

// Driver code
int main()
{
    // Create a Priority Queue
    // 7>4>5>6
    Node* pq = newNode(4, 1);
    pq = push(pq, 5, 2);
    pq = push(pq, 6, 3);
    pq = push(pq, 7, 0);

    while (!isEmpty(pq)) {
        printf("%d ", peek(pq));
        pq = pop(pq);
    }

    return 0;
}
```

Applications of Priority Queue



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- **CPU Scheduling:**
Priority queues are used in operating systems to manage process scheduling, where processes with higher priority are executed before others.
- **Graph Algorithms:**
Dijkstra's shortest path algorithm and Prim's minimum spanning tree algorithm use priority queues to select the next node with the smallest weight.
- **EventDriven Simulation:**
In simulations, priority queues are used to manage events, scheduling the next event based on the time of occurrence.
- **Huffman Encoding:**
Priority queues help build Huffman trees in data compression by always combining the two least frequent nodes.
- **Load Balancing:**
Used in distributed systems to allocate tasks to servers, prioritizing servers with lighter Loads.
- **AI and Search Algorithms:**
Priority queues are utilized in A* search algorithms and other pathfinding algorithms to explore the most promising nodes first.
- **Network Traffic Management:**
Used to prioritize network packets, ensuring highpriority packets (like realtime video/audio) are transmitted first.
- **Task Scheduling in RealTime Systems:**
In realtime systems, tasks are scheduled based on their urgency using priority queues.
- **Operating System Interrupt Handling:**
Interrupts are handled in order of priority, ensuring critical operations are handled first.
- **Data Stream Processing:**



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In systems that process large data streams, priority queues help manage which elements should be processed next, prioritizing urgent or important data.